Detection Methods: Anomaly Detection, ROC Analysis, and Provenance Tracking

Detection Methods: Anomaly Detection, ROC Analysis, and Provenance Tracking I

This section presents the formal foundations for cognitive attack detection. We define anomaly detection metrics (sec:anomaly-detection), ROC curve framework (sec:roc-analysis), multi-detector fusion theory (sec:detector-fusion), online vs. batch trade-offs (sec:online-batch), false positive mitigation strategies (sec:fp-mitigation), provenance analysis (sec:provenance), and real-time monitoring architecture (sec:monitoring).

***Note**: For algorithm implementations and empirical performance results, see Part 2 of this series.*

Anomaly Detection I

Cognitive Drift Scoring

Definition (Drift Score)

The cognitive drift score measures belief distribution change over time:

$$S_{\mathsf{drift}}(t) = D_{\mathsf{KL}}(\mathcal{B}_i^t \| \mathcal{B}_i^{t-w}) + \lambda \cdot \max_{\phi} |\Delta \mathcal{B}_i(\phi)| \tag{1}$$

Table 1: Drift score components and detection targets.

Component	Weight	Detection Target
KL divergence	1.0	Gradual distribution shift
Max delta	λ	Sudden belief injection

[Drift Detection Threshold] For normally distributed baseline drift, the threshold $\theta = \mu_{\mathsf{baseline}} + k \cdot \sigma_{\mathsf{baseline}}$ with $k = 3$ provides 99.7% confidence under the null hypothesis of no attack.

Anomaly Detection II

Behavioral Deviation

Definition (Deviation Score)

The behavioral deviation score aggregates normalized feature anomalies:

$$S_{\mathsf{dev}}(a_i, t) = \sum_{k=1}^{K} w_k \cdot \frac{|f_k(\sigma_i^t) - \mu_k|}{\sigma_k} \tag{2}$$

where f_k are feature extractors and (w_k, μ_k, σ_k) are learned parameters.

Anomaly Detection III

Ensemble Detection

Definition (Ensemble Detector)

Combines multiple detector scores via learned fusion:

$$P(\text{attack} \mid \mathbf{S}) = \sigma\left(\sum_d w_d \cdot S_d - b\right) \tag{3}$$

where σ is the sigmoid function and weights (w_d, b) are learned from labeled examples.

ROC Curve Analysis I

ROC Curve Analysis II

Receiver Operating Characteristic Framework

Definition (ROC Curve)

For detector D with threshold τ:

$$\text{ROC}(D) = \{(\text{FPR}(\tau), \text{TPR}(\tau)) : \tau \in [0, 1]\} \tag{4}$$

where the rates are defined as:

$$\text{TPR}(\tau) = P(D(x) > \tau \mid x \in \mathcal{A}_{\text{attack}}) \tag{5}$$

$$\text{FPR}(\tau) = P(D(x) > \tau \mid x \in \mathcal{X}_{\text{benign}}) \tag{6}$$

Definition (Area Under Curve)

$$\text{AUC}(D) = \int_0^1 \text{TPR}(\text{FPR}^{-1}(t))\, dt \tag{7}$$

Table 2: AUC interpretation scale.

Multi-Detector Fusion I

Multi-Detector Fusion II

Fusion Strategies

Definition (Score-Level Fusion)

Weighted average of detector outputs:

$$S_{\text{fused}} = \sum_{i=1}^{k} w_i \cdot S_i, \quad \sum_i w_i = 1 \tag{10}$$

Definition (Decision-Level Fusion)

Quorum voting on binary decisions:

$$D_{\text{fused}}(x) = \mathbb{1}\left[\sum_{i=1}^{k} \mathbb{1}[D_i(x) > \tau_i] \geq q\right] \tag{11}$$

Definition (Learned Fusion)

Neural network combining scores:

Online vs. Batch Detection I

Comparison Framework

Table 3: Online vs. batch detection trade-offs.

Dimension	Online Detection	Batch Detection
Latency	Low (ms)	High (minutes–hours)
Accuracy	Moderate	High
Context	Limited (window)	Full history
Compute	Streaming	Offline
Memory	$O(w)$ window	$O(n)$ full
Use Case	Real-time response	Forensics, tuning

Online vs. Batch Detection II

Streaming Detector Model

Definition (Streaming Detector)

Processes messages in real-time with bounded memory:

$$D_{\mathsf{online}}(m_t) = f(m_t, \mathsf{state}_{t-1}) \tag{15}$$

$$\mathsf{state}_t = g(\mathsf{state}_{t-1}, m_t) \tag{16}$$

Online vs. Batch Detection III

Hybrid Detection Architecture

Definition (Hybrid Detection System)

Combines online and batch detection via feedback loop:

$$\text{Online Path}: m \xrightarrow{\text{filter}} s \xrightarrow{\text{decide}} r \xrightarrow{\text{log}} H \tag{17}$$

$$\text{Batch Path}: H \xrightarrow{\text{analyze}} \text{patterns} \xrightarrow{\text{update}} \text{filters} \tag{18}$$

False Positive Mitigation I

Strategy 1: Confirmation Cascade

Definition (Confirmation Cascade)

Multi-stage verification before alerting:

$$\text{Action}(\text{confidence}) = \begin{cases} \text{suppress} & \text{if } c < C_{\text{low}} \\ \text{stage-2} & \text{if } c \in [C_{\text{low}}, C_{\text{high}}) \\ \text{stage-3} & \text{if } c \geq C_{\text{high}} \end{cases} \tag{19}$$

Theorem (Cascade FPR Reduction)

For a multi-stage cascade:

$$\mathit{FPR}_{\mathit{cascade}} = \mathit{FPR}_1 \cdot P(\mathit{confirm}_2 \mid \mathit{FP}_1) \cdot P(\mathit{confirm}_3 \mid \mathit{FP}_2) \tag{20}$$

False Positive Mitigation II

Strategy 2: Temporal Smoothing

Definition (Smoothed Detection)

Apply exponential smoothing to scores:

$$\hat{S}_t = \alpha \cdot S_t + (1 - \alpha) \cdot \hat{S}_{t-1} \tag{21}$$

Definition (Burst Suppression)

Require sustained anomaly over window w:

$$\text{Alert if } \frac{1}{w} \sum_{i=t-w+1}^{t} \mathbb{1}[S_i > \tau] > p_{\text{sustained}} \tag{22}$$

False Positive Mitigation III

Strategy 3: Contextual Whitelisting

Definition (Context-Aware Whitelist)

$$\text{Suppress}(\text{alert}) \iff \text{context}(\text{alert}) \in \mathcal{W}_{\text{known}} \tag{23}$$

Strategy 4: Cost-Sensitive Thresholding

Definition (Cost-Sensitive Threshold)

Optimize for total cost rather than accuracy:

$$\tau^* =_{\tau} \left[C_{\text{FP}} \cdot \text{FPR}(\tau) + C_{\text{FN}} \cdot \text{FNR}(\tau)\right] \tag{24}$$

Provenance Analysis I

Provenance Analysis II

Information Flow Tracking

Definition (Taint Label)

Each belief carries provenance tags:

$$\text{taint}(\phi) = \{(\text{source}, \text{timestamp}, \text{confidence})\} \tag{25}$$

Definition (Taint Propagation)

$$\text{taint}(\phi_{\text{derived}}) = \bigcup_{\psi \in \text{premises}(\phi_{\text{derived}})} \text{taint}(\psi) \tag{26}$$

Table 4: Taint categories with trust levels.

Category	Trust Level	Example
SYSTEM_VERIFIED	1.0	Hardcoded facts
PRINCIPAL_INPUT	0.9	Direct user commands
AGENT_INTERNAL	0.8	Agent's own reasoning

Real-Time Monitoring I

Alert Aggregation

Definition (Alert Aggregation)

Prevent alert fatigue through correlation:

$$\text{Severity} = \begin{cases} \textsc{critical} & \text{if } |\text{alerts}| > n_{\text{critical}} \text{ in window } w \\ \textsc{warning} & \text{if } |\text{alerts}| > n_{\text{warning}} \text{ in window } w \\ \textsc{info} & \text{otherwise} \end{cases} \tag{29}$$

Real-Time Monitoring II

Response Escalation

Table 6: Response escalation levels.

Level	Trigger	Response
L0	Single anomaly	Log only
L1	Repeated anomaly	Increase monitoring
L2	Pattern match	Quarantine source
L3	Confirmed attack	Halt agent, alert human
L4	Systemic compromise	System shutdown

Real-Time Monitoring III

Empirical Validation

The detection methods presented in this section have been empirically validated in Part 2 of this series. Key results include:

ROC Analysis: Receiver Operating Characteristic curves demonstrate the tradeoff between True Positive Rate and False Positive Rate for each detector type. The ensemble achieves AUC > 0.84, with individual mechanisms ranging from 0.74 (Belief Sandbox) to 0.81 (Tripwire Monitor). See Part 2, §{4} for detailed ROC curves and confidence intervals.

Detection Performance by Attack Type: Detection rates vary across the five adversary classes (Ω_1–Ω_5). The Cognitive Firewall excels at Ω_1 (external) attacks while Tripwires and Invariants provide stronger coverage for Ω_3 (compromised agent) and Ω_4 (inter-agent) attacks. See Part 2, §{5} for the complete detection matrix.

False Positive Mitigation: The confirmation cascade, temporal smoothing, and contextual whitelisting strategies reduce false positive rates by $> 80\%$ while maintaining $> 90\%$ true positive rates. See Part 2, §{5.4} for quantitative analysis of each